Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for

Prevention of Early Onset Group B Streptococcal Sepsis

Eastern and Northern Services

Summary

This guideline outlines the process for diagnosis, management and treatment of infants and women with maternal GBS infection/ colonisation.

Key Points

See flowchart in Appendix 2 on page 9

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1. INDICATIONS FOR MATERNAL INTRAPARTUM ANTIBIOTIC PROPHYLAXIS (IAP)

1.1 The following women should be offered IAP- if identified antenatally place a GBS sticker in the hand-held notes:

- Women with positive vaginal swab or urine culture for GBS at any stage in the **current** pregnancy, unless delivery is by planned elective caesarean section at term with intact membranes in the absence of labour.
- Women who have had a previous infant with early onset Group B Streptococcal sepsis, unless delivery is by planned elective caesarean section at term with intact membranes in the absence of labour.
- Women who had GBS detected in a **previous** pregnancy and have declined or missed screening at 35-37/40 (as the risk of carriage in this pregnancy is 50%), unless delivery is by planned elective caesarean section at term with intact membranes in the absence of labour.
- Women in confirmed preterm labour.
- Women who are induced for preterm pre-labour rupture of membranes regardless of their GBS status.

1.2 Patient information – The RCOG have compiled a patient information leaflet than can be access via this hyperlink: https://www.rcog.org.uk/globalassets/documents/patients/patient-information-leaflets/pregnancy/pi-gbs-pregnancy-newborn.pdf

1.3 Antibiotic regime:

Women who require IAP should be given the first dose as soon as possible (preferably within one hour) and continue until delivery.

	Initial Dose	Maintenance Therapy until Delivery	Comments
First Line	Benzyl penicillin 3g IV*	Benzyl penicillin 1.5gIV**every 4 hours	Maximum rate 300mg/min
Mild Penicillin Allergy	Cefuroxime 1.5g	Cefuroxime 750mg every 8 hours	Absence of anaphylaxis,angioedema, respiratory distree or urticaria
Moderate/Severe Penicillin Allergy	Vancomycin 1g IV	Vancomycin 1g IV every 12 hours	Maximum rate 10mg/min - therefore infuse over at least 100mins to avoid side effects

*3g = two x 1.2g vials and one x 600mg vial

**1.5g = one 1.2g vial and half x 600mg vial

1.4 The Table in Appendix 1 gives national estimates risk of EOGBS disease in the presence of individual antenatal risk factors, with and without Intrapartum Antibiotics. Decision should be made in consultation with obstetrician regarding use of IAP antibiotics in cases other than those in point 1.1

2. INDICATIONS FOR ANTE NATAL ANTIBIOTICS

2.1 Asymptomatic women with Group B Strep on MSU should receive a course or oral antibiotics. The antibiotic of choice is oral amoxicillin or oral cephalexin or nitrofurantoin in penicillin allergic mothers. Women with symptoms of a UTI will generally have already been treated empirically (according to guideline), but if not, should receive a course of appropriate oral antibiotics.

2.2 In women with asymptomatic GBS bacteruria, do NOT send a test of cure urine sample

2.3 Ensure a Group B Strep is noted on the pregnancy story board – this should automatically be added with a positive culture, It can also be added to the problem list.

2.4 Advise the woman she will need IV antibiotics in labour and will be asked to deliver in the RD&E Labour Ward or, if no other risk factors, the RD&E Low Risk Birth Centre.

2.5 Women who receive IAP may use the birthing pool for labour and birth (see Water birth guideline).

2.6 Antenatal treatment is not recommended for GBS cultured from a vaginal swab.

3. ANTENATAL SCREENING

3.1 Universal bacteriological screening is not recommended and a maternal request is not an indication for bacterial screening

3.2 In women who in a previous pregnancy were GBS positive, the likelihood of maternal GBS carriage in their current pregnancy is 50 %. RCOG advises offering screening in this population - screening is performed by low vaginal then rectal swab (Appendix 3). Ensure the request form states GBS screening only.

4. ANTENATAL IDENTIFICATION OF GBS

4.1 Following identification of GBS in current pregnancy either by MSU, LVS or screening swabs, the woman must be informed via letter (Appendix 4) and a GBS should be flagged on the storyboard. A link to the RCOG patient information leaflet is highlighted in the letter.

5. ADDITIONAL MANAGEMENT OF WOMEN WITH GBS IN CURRENT PREGNANCY OR GBS IN PREVIOUS PREGNANCY THAT ARE UNSCREENED (see Appendix 3)

5.1 Immediate IOL and IAP is advised in all women with pre labour rupture of membranes (PROM) at 34 weeks or more. Management in this way reduces the risk of early onset neonatal sepsis from 15.2% with expectant management to 1.8% (PPROMEXIL trial)

5.2 In women with **preterm** PROM less than 34 weeks, routine antenatal Erythromycin will be given (see Pre-term Pre-labour Rupture of Membranes guideline). Immediate IOL is not required but IAP should be commenced when in spontaneous labour or when induction of labour occurs at appropriate gestation.

6. DEFINITION OF ADEQUATE IAP

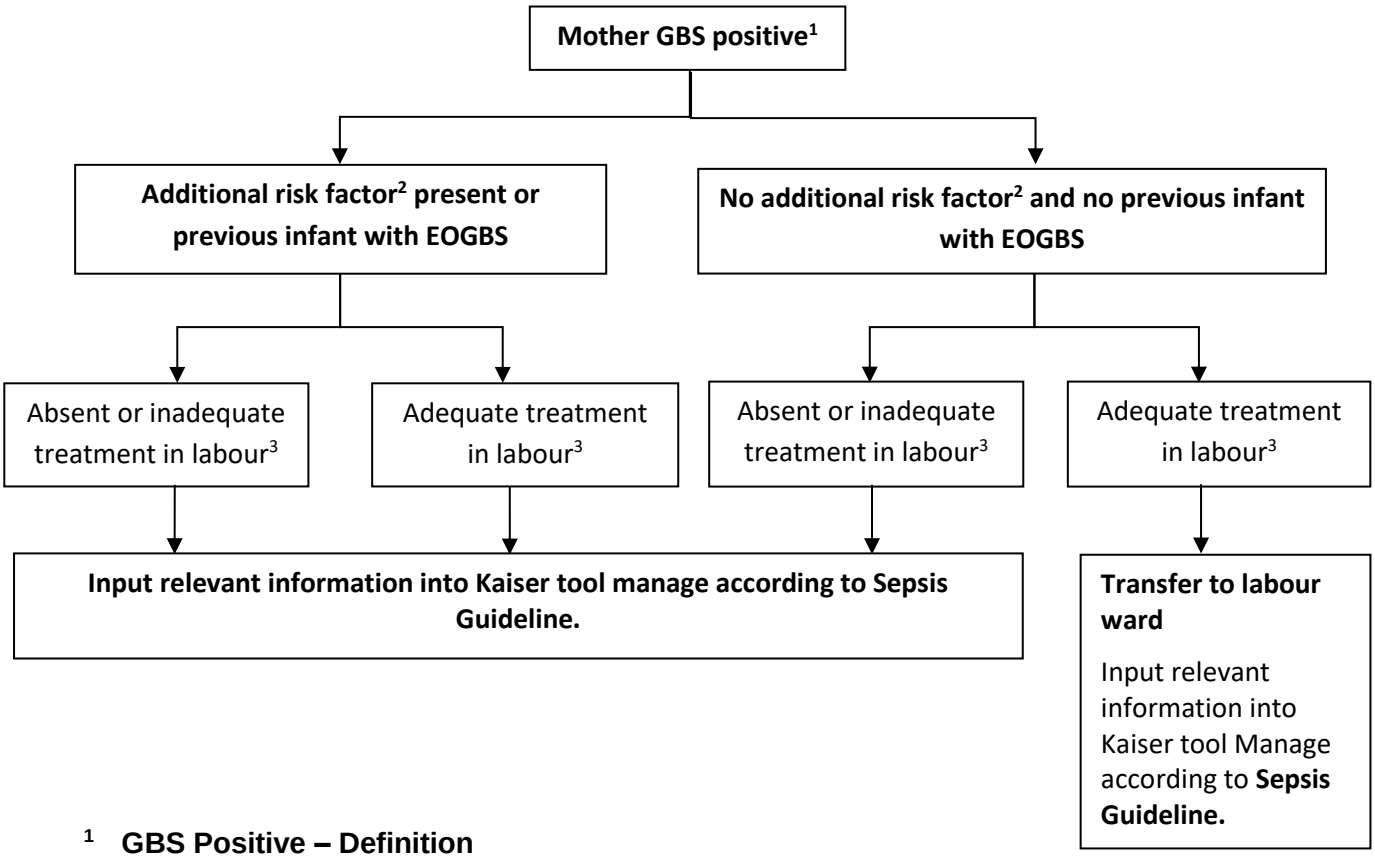
6.1 Antibiotics given at least 4 hours prior to delivery. Doses given should be Benzylpenicillin 3g IV initially then 1.5g IV at 4 hourly intervals until delivery (in cases of Penicillin allergy Vancomycin 1g 12 hourly).

6.2 Mothers who deliver by caesarean section at term with intact membranes in the absence of labour do not need to receive IAP even if found to be GBS positive in present pregnancy and their baby does not need a Kaiser Assessment.

6.3 If chorioamnionitis is suspected, broad-spectrum antibiotic therapy including an agent active against GBS should replace GBS-specific prophylaxis. If in doubt, discuss with on call microbiologist.

7. MANAGEMENT OF INFANTS WITH MATERNAL GBS INFECTION/COLONISATION

All infants of GBS positive mothers (except those who deliver at term by caesarean section with intact membranes in the absence of labour) regardless of whether the mothers have had IV Antibiotics should be entered into the Kaiser Calculator.



[1] **GBS Positive – Definition**

- "GBS positive" is defined as a positive GBS culture from any site at any stage in the current pregnancy or in a previous pregnancy (if screening declined or not performed **in the current pregnancy).**
- In addition, women who have had a previous infant with EOGBS sepsis should be regarded as GBS positive irrespective of any culture results in current pregnancy.

[2] **Any of these risk factors the infant the mother's details should be inputted into the Kaiser calculator: https://neonatalsepsiscalculator.kaiserpermanente.org/**

- Mother with GBS in urine at any time during current pregnancy
- Preterm (<37 weeks gestation) labour
- Preterm rupture of membranes (with or without labour)
- Prolonged rupture of membranes (PROM) >24hours with or without labour
- Maternal pyrexia (>37.8°C) during labour

[3] **See "Definition of adequate IAP" above.**

[4] **Screening and treatment of infants**

- FBC / CRP / Blood cultures
- If infant requires antibiotics according the Kaiser calculation then prescribe - IV benzylpenicillin and gentamicin for minimum of 36 hours after screening.

[5] **Observations**

The observations are commonly called Newborn Early Warning Trigger and Track (NEWTT) observations and should include:

- General Behaviour which may include muscle tone and feeding
- Skin colour and perfusion
- Grunting
- Temperature
- Heart rate
- Respiratory rate
- The observations should be documented on a NEWTT chart

Observations that require action:

In the event of 1 amber NEWTT – escalate concern to the Labour Ward coordinator and review in 30 mins.
In the event of 2 amber or 1 red NEWTT – escalate to a neonatal doctor immediately by bleeping 745.

8. REFERENCES

Guideline no. 36 – 'Prevention of Early Onset Neonatal Group B Streptococcal Disease'; Royal College of Obstetricians and Gynaecologists (September 2017)

National Institute for Clinical Excellence. Intrapartum Care: Care of Health Women and their babies during childbirth. NICE. London.

Newborn Early Warning Trigger and Track (NEWTT) - A Framework for Practice;

The British Association of Perinatal Medicine (April 2015).

9. MONITORING COMPLIANCE WITH THIS GUIDELINE

9.1 Any concern or non-compliance with this guideline that is identified through the investigation of clinical incidents, claims or complaints will be reviewed as per the Trust Policies regarding Incidents, Claims and Complaints, and may result in an audit and/or amendment to the guideline.

9.2 Relevant Policies:

- Incident reporting, analysing, investigating and learning policy and procedures
- Claims Management Policy and Procedure
- Policy and Procedure for the Management of Complaints, Concerns, Comments and Compliments

10. ASSOCIATED CLINICAL GUIDELINES

- Water birth
- Pre-term Pre-labour Rupture of Membranes guideline
- Sepsis Guideline

11. PUBLICATION DETAILS

Author of Clinical Guideline	Lead Consultant for Neonatal Services and Labour Ward Lead Consultant
Division/Department responsible for Clinical Guideline	Specialist Services/Child and Women's Health/Neonatal & Maternity
Contact details	6629
Version number	1.0 Joint services guideline
Replaces version number	v4.2 East and v5.0 North
Date written	July 2003, revised July 2020, updated September 2023
Approving body and date approved	Maternity Governance Chair's Action 09/09/2024 NNU Governance 09/10/2024
Review date	09/06/2027
Expiry date	09/09/2027
Date document becomes live	09/10/2024

APPENDIX 1 - Royal College of Obstetricians and Gynaecologists (RCOG) Green-top Guideline No. 36 (July 2012) – The Prevention of Early-onset Neonatal Group B Streptococcal Disease, Appendix I, p.10

Estimates of the risk of EOGBS disease in the presence of individual antenatal risk factors, with and without IAP.

These estimates are based on an incidence of EOGBS disease in the UK of 0.5/1000, which is likely to be the minimum incidence.

Risk factor	Risk of EOGBS disease if IAP not given	Risk of EOGBS disease if full IAP given	Risk of death from EOGBS disease if IAP not given	Risk of death from EOGBS disease if full IAP given
Intrapartum fever (>38°C)	1:189	1:943	1:1783	1:8915
Prolonged rupture of membranes (>18 hours) at term	1:556	1:2777	1:9754	1:48,772
Prematurity (<37+0 weeks of gestation)	1:435	1:2173	1:2377	1:11,885
Prematurity (<35+0 weeks of gestation)	1:357	1:1786	1:1566	1:7829
Positive GBS swab in a previous pregnancy	1:1105	1:5525	1:10,424	1:52,122
Positive GBS swab in current pregnancy	1:434	1:2170	1:4094	1:20,471

EOGBS = early onset group B streptococcus; IAP= intrapartum antibiotic prophylaxis.

The assumptions on which the figures in the table above are based are as follows:

- Live birth rate in the UK in 2008: 793,388
  - 1.9% intrapartum fever >38°C
  - 9.4% PROM at term
  - 7.9% <37 weeks of gestation
  - 4.0% <35 weeks of gestation

- Prevalence of maternal risk factors in infants with EOGBS disease:
  - 19.9% intrapartum fever >38°C
  - 34% PROM at term
  - 37% <37 weeks of gestation
  - 22% <35 weeks of gestation

- Incidence of EOGBS in the UK: 0.5/1000

- Mortality of EOGBS in the UK is:
  - 10.6% overall
  - 18.3% <37 weeks of gestation
  - 22.8% <35 weeks of gestation
  - 5.7% >37 weeks of gestation

- 80% effectiveness of IAP in preventing EOGBS

It should be noted that GBS bacteriuria is a risk factor for neonatal disease but the magnitude of risk cannot be quantified.

APPENDIX 2 - GROUP B STREPTOCOCCAL SEPSIS FLOWCHART

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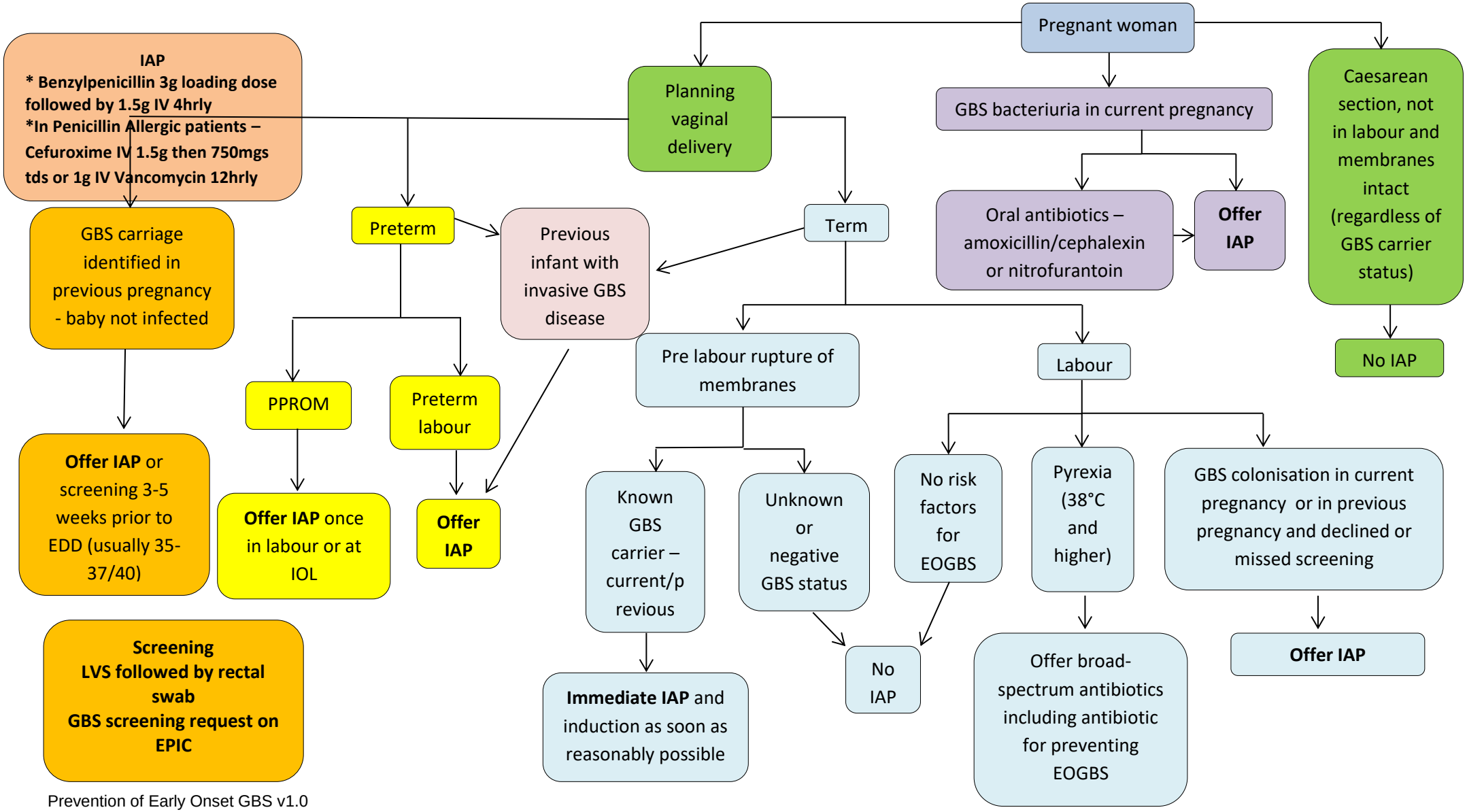
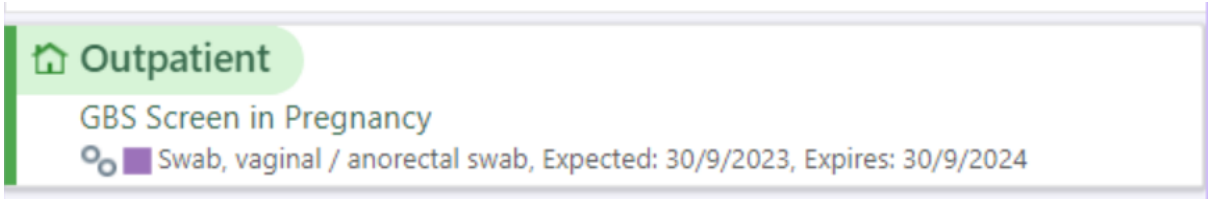


APPENDIX 3 – BACTERIOLOGICAL TESTING FOR GBS

- Universal bacteriological screening is not recommended
- Women who have previously had GBS detected in pregnancy (with an unaffected baby) have a 50% chance of carriage in their current pregnancy. These women should be offered IAP, or bacteriological testing in late pregnancy and then offered IAP if still positive.
- Bacteriological testing should ideally be carried out at 35-37 weeks or 3-5 weeks prior to anticipated delivery date
- Bacteriological testing for GBS – low vaginal followed by anorectal swab and request a GBS screen.

How to test for GBS Carriage

- Explain to the woman the importance of both vaginal and anorectal swab
- Obtain consent
- Perform low vaginal swab **first** and then place the same swab into the anorectum – some women may prefer to swab themselves
- Ensure that the order is for a GBS screen – some women may decline anorectal swab and this should be documented on the form



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APPENDIX 4 - GBS LETTER TO PATIENT

Fetal Maternal Assessment Unit
Centre for Women's Health
Royal Devon & Exeter Foundation NHS Trust
Barrack Road
Exeter
EX2 5DW

Tel 01392 406540

Dear

Re: Group B Streptococcus result
Your recent urine/swab result has shown that you carry Group B Streptococcus (GBS). This is a common bacterial infection which is found intermittently in the vagina of a quarter of women.

Here is a link to an information leaflet for further information regarding the significance of GBS in pregnancy and the treatment that is recommended -

https://www.rcog.org.uk/globalassets/documents/patients/patient-information-leaflets/pregnancy/pi-gbs-pregnancy-newborn.pdf

Your own midwife will have been informed.

Yours sincerely

Midwife

Fetal Maternal Assessment Unit
01392 406540